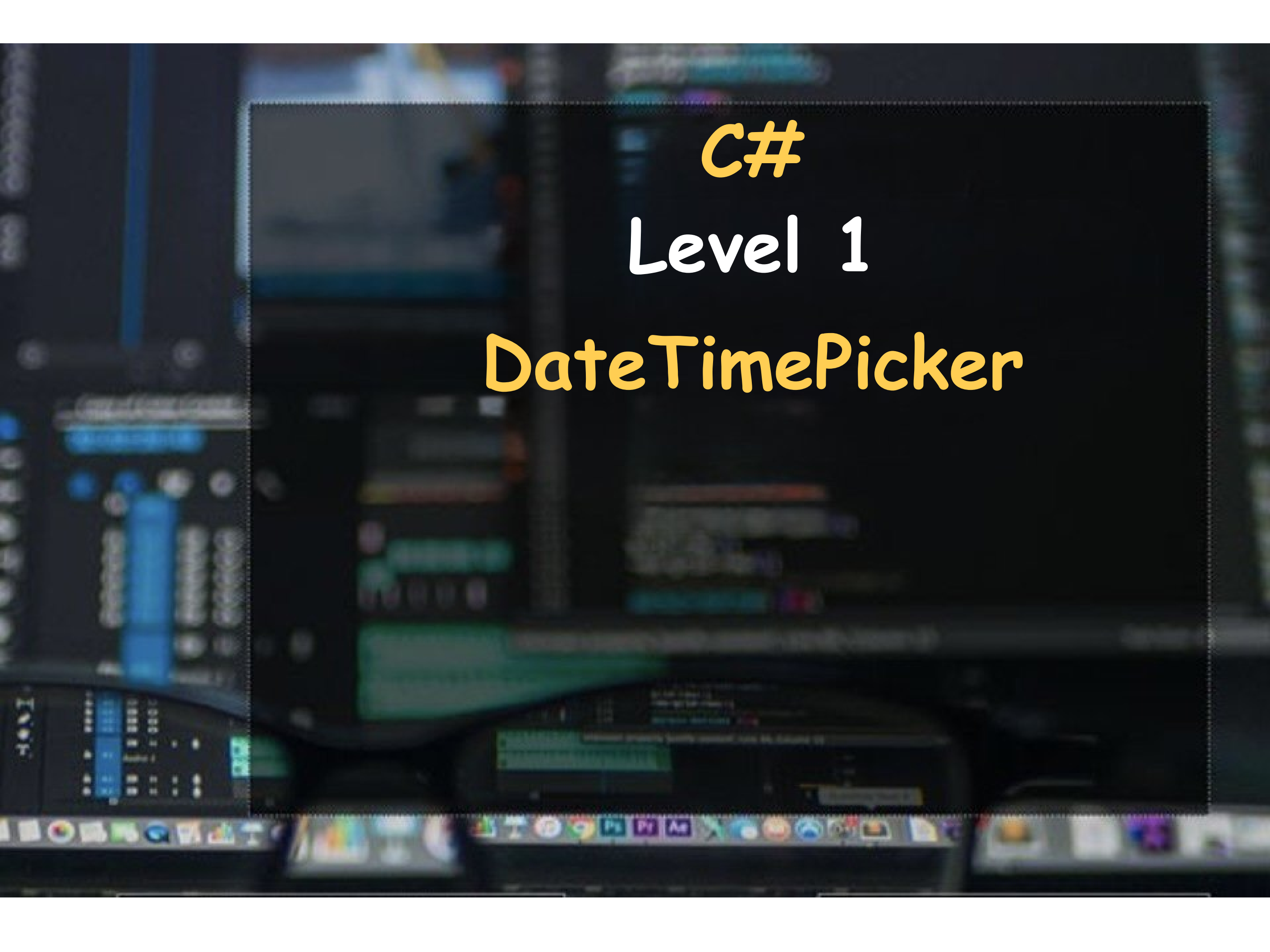




C#

Level 1

DateTimePicker



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Definition

DateTimePicker

-  Windows Forms control used to select a date and time.
-  The selected value is accessed through the `Value` property.
-  The value can be displayed using standard or custom date/time formats.

Explanation

Value

-  ``dateTimePicker1.Value`` returns the selected ``DateTime``.

-  The value can be converted to different string formats.

ValueChanged

-  Occurs whenever the selected date or time changes.

Date Formatting

-  ``ToShortDateString()`` displays a culture-dependent short date.

-  ``ToLongDateString()`` displays a culture-dependent long date.

-  ``ToString("format")`` allows a custom date format.

Procedure

1. Add a DateTimePicker to the Windows Form.
2. Select a date from the calendar.
3. Access the selected value using `Value`.
4. Convert the value using a standard or custom format.
5. Use `ValueChanged` when the selected value changes.
6. Display the formatted result in a Label or MessageBox.

💡 Example

```
private void button1_Click(object sender, EventArgs e)
{
    MessageBox.Show(dateTimePicker1.Value.ToShortDateString());
}
```

◆ Custom format:

```
dateTimePicker1.Value.ToString("dd-MMM-yyyy")
```

◆ `dd` → two-digit day.

◆ `MMM` → abbreviated month.

◆ `yyyy` → four-digit year.

➡ Result

If the selected date is September 10, 2026:

`ToShortDateString()`

→ 2026-09-10 or 09/10/2026 depending on culture.

`ToLongDateString()`

→ Thursday, September 10, 2026 depending on culture.

`ToString("dd-MMM-yyyy")`

→ 10-Sep-2026

Note

◆ Important DateTimePicker properties:

- Value → gets or sets the selected date and time.
- MinDate → sets the earliest selectable date.
- MaxDate → sets the latest selectable date.
- Format → controls the display format.
- CustomFormat → defines a custom format when Format is Custom.
- ShowUpDown → displays up/down controls instead of the calendar.
- ShowCheckBox → displays a checkbox.
- Checked → indicates whether the selected value is enabled when

ShowCheckBox is used.

◆ Common format characters:

- d → day without leading zero.
- dd → two-digit day.
- ddd → abbreviated day name.
- dddd → full day name.
- M → month without leading zero.
- MM → two-digit month.
- MMM → abbreviated month name.
- MMMM → full month name.
- yy → two-digit year.
- yyyy → four-digit year.



Comparison



`ToShortDateString()`

- ✓ Short date format.
- ✓ Uses the current culture.



`ToLongDateString()`

- ✓ Long date format.
- ✓ Uses the current culture.



`ToString("format")`

- ✓ Custom date/time format.
- ✓ Gives more control over the displayed structure.

⊘ Warning

- ◆ `ToString()` and `ToLongDateString()` depend on the system's culture settings.
- ◆ Use custom formats when a specific output structure is required.
- ◆ `MM` means month, while `mm` means minutes.

✦ Summary

- ✓ `DateTimePicker` allows the user to select a date and time.
- ✓ ``Value`` → gets the selected `DateTime`.
- ✓ ``ValueChanged`` → occurs when the selected value changes.
- ✓ ``ToShortDateString()`` → short culture-dependent date.
- ✓ ``ToLongDateString()`` → long culture-dependent date.
- ✓ ``ToString("format")`` → custom date/time format.
- ✓ ``MinDate`` / ``MaxDate`` → limit selectable dates.
- ✓ ``Format`` / ``CustomFormat`` → control the displayed format.

Thank you 😊